QuESO Official Manual

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Warning

This documentation is under active development.

CHAPTER

ONE

INSTALLATION

You can install the latest version of this package from Codeberg by

```
pip install git+https://codeberg.com/SharkSorceress/QuESO.git
```

Once we have a proper version, we plan to submit our package to PyPi as `queso-cluster`.

CHAPTER

TWO

API REFERENCE

2.1 writer.py

exportFITS(*spectralDataset*, *labelLine*, *fname*)

2.2 aux.py

_gen_dataID(*Input*)

pick_jth_label(*labelLst*, *j*)

density_2channel(*x*, *y*, *dy*, *xsize*, *top*, *bottom*)

density_hist2d(*data*, *dy*, *top*, *bottom*)

close_factors(*number*)

almost_factors(*number*)

common_elements(*ar1*, *ar2*, *ar3*)

2.3 base.py

normZ(*dataSquare*)

normMaximum(*dataSquare*)

normContinuum(*dataSquare*, *continuumIndx*)

concatSpectra(*dataSquareLst*)

numba_histogram(*a*, *bins*, *lim*)

rotateArray(*image*, *turns*)

get_bin_edges(*bins*, *lim*)

compute_bin(*x*, *bin_edges*)

np_gradient(*f*)

minimize(*data*, *decisions*, *size*)

maximize(*data*, *decisions*, *size*)

np_all_axis0(*x*)

np_all_axis1(*x*)

similarityMetric(*x*, *y*, *type='dist'*, *ref=0*)

curvature(*y*)

startMax(*data*, *k*, *decisions*)

startPlusPlus(*data*, *k*, *decisions*)

_calcMoment(*waveAxis*, *ii*, *jj*, *lineCore*, *dataCube*, *order*, *ref*, *counter=0*)

_calcFeatureDensity(*data*, *converge*, *zindx*, *func1*)

_calcOptimization(*k*, *data*, *decision*, *threshold*)

_calcElbowEntry(*data*, *labels*)

_calcVarianceScore(*data*)

_criteriaInertiaScore(*score*)

_calcInertiaScore(*dataSquare*, *labelLine*)

_criteriaSilhouetteScore(*score*)

_calcSilhouetteScore(*dataSquare*, *labelLine*)

_calcSingleSilhouetteScore(*data*, *labels*, *lab*)

_calcCHindex(*data*, *labels*)

labelGluer(*labels*)

labelReorder(*labels*)

_calcQuiescentFrame(*spectralData*, *spectralParams*, *contIndxs*, *progress=None*)

_calcDynamicFrame(*spectralData*, *dynamicScanNum*, *progress=None*, *delta=0*)

2.4 approach.py

__init__(*self*, *config*, *catalogName*, *instrumentObj*)

__getattr__(*self*, *name*)

cluster(self,prepSquare,maskLine

__init__(*self*, *catalogBase*)

clustering(*self*, *prepCube*, *tlst*, *groups*, *intrinsicSquare=None*)

Detail
low temporal resolution clustering

2.5 logg.py

loggTimer(*func*)

wrapper(**args*, ***kwargs*)

logg(*tag*, *val=None*, *_time=None*, *_log=None*)

duration_string(*dur*)

2.6 aia.py

delayAIA(*fname*, *epochDev*)

2.7 style.py

cbar_bounds(*bounds*)

_genColorPallet(*n*)

rainbow_cmap(*nrange*, *discrete=False*, *nan=False*)

cmap_discretize(*cmap*, *N*)

__init__(*self*, *spaceInfo*, *deltas*)

_mapGen(*self*, *fig*, *pos*, *arr*, *flareContour=None*, *timeAxis=None*, ***kwargsDict*)

2.8 multiline.py

2.9 products.py

__init__(*self*, *epochObj*, *optLabels*)

figure03(*self*)

figure04_template(*self*)

spectralEntry(*self*, *ax*, *indx*, *color*, *wavelambda*, *extent*, *scores=None*)

2.10 loader.py

__init__(*self*, *data*, *home*, *fig*)

_loadEventConfig(*self*, *eventRunnerFname*, *args*)

__init__(*self*, *dataPath*)

vispLoad(*self*, *stokes=0*)

irisLoad(*self*)

fissLoad(*self*, *labels*)

__init__(*self*, *config1*, *config2*)

visp2visp(*self*)

fiss2fiss(*self*)

__init__(*self*, *dataPath*, *labels*)

__init__(*self*, *dataPath*, *stokes=0*)

__init__(*self*, *inputLst*, *runIndx*)

loadSource(*self*, *eventInput*)

__init__(*self*, *srcInput*)

__init__(*self*, *runnerInput*)

__init__(*self*, *fname*, *eventIndx*, *runIndx*)

_load(*self*, *fname*)

2.11 base.py

_mainIntrinsic(*config*, *prepSquare*, *lineIndx*, *intrinsicSkip=False*)

_mainOptimization(*prepSquare*, *labelLine*, *kLst=None*, *stageMax=2*)

2.12 base.py

_runOptimalKSearch(*dataSquare*, *funcLst*, *checkLst*)

runPrep(*dataSquare*, *norm='continuum'*, *keepI0=None*, *maskLine=None*, *quSquare=None*, ***kwargs*)

_runIntrinsic(*nbins*, *data*, *edgeOverride=None*)

runStart(*k*, *data*, *start='max'*)

_runOptimization(*k*, *sub_data*, *converge*)

_findOptimalK(*dataSquare*, *funcLst*, *criteriaLst*, *converge=1e-6*)

_runLabelSort(*dataSquare*, *labelLine*)